Agenda:

1. Course outcome, evaluation
2. Material review,
3. Neural Networks
4. Feed Forward Propagation

Course Outcome:

1. Explain Deep Learning terminologies of an AI ecosystem.
2. Apply advanced Natural Language Processing techniques for Sentiment Analysis, Entity Recognition, and Text classification.
3. Build Deep Learning models using frameworks such as Tensorflow and Keras for NLP (Natural Language Processing) and audio analysis.
4. Build, train and evaluate RNN
5. Build and train different type of Autoencoders
6. …

Course Evaluation:

Course CSI

Learning through discussion, engagements and active participation

Assignment's presentation

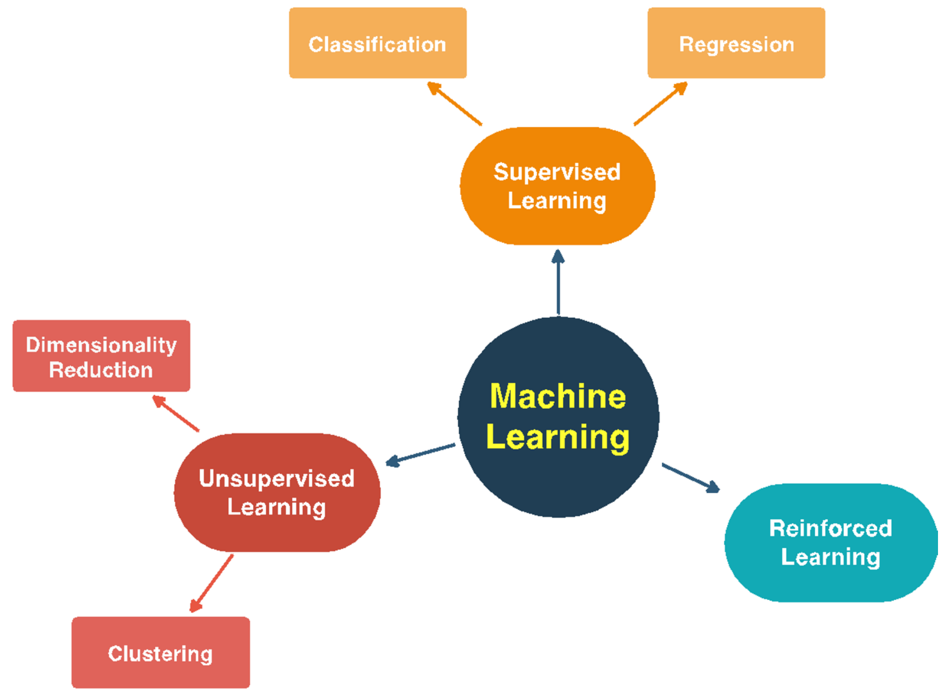
Discuss coding strategies, plan of attacks, and solutions…

Machine Learning subcategories:

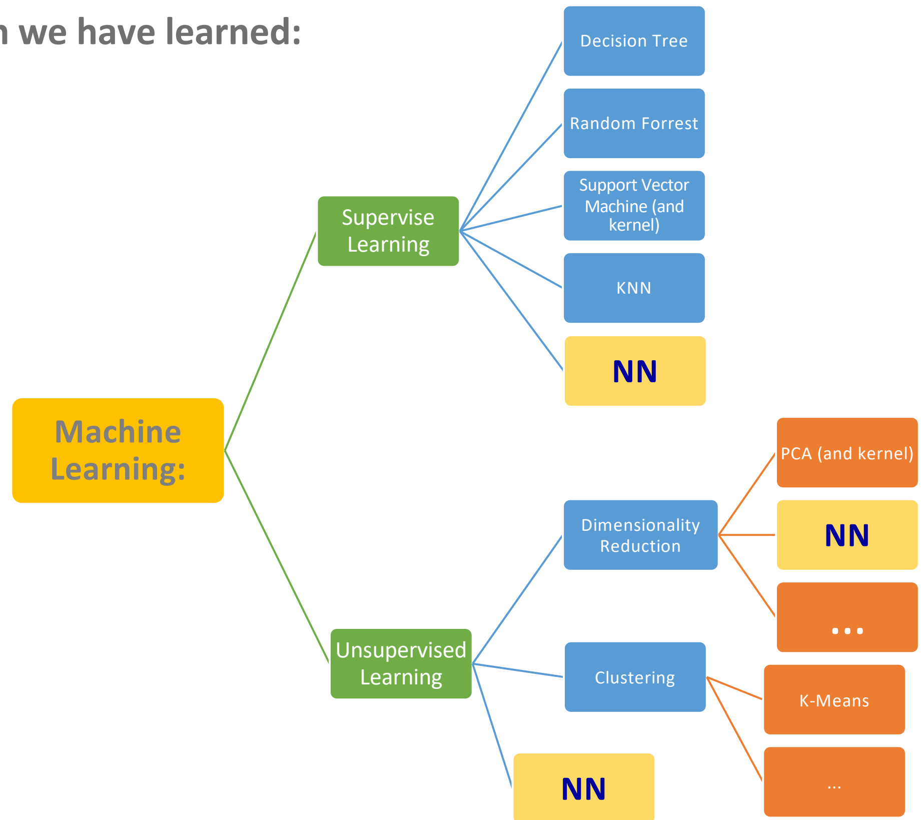
I. Supervised Learning
II. Un-supervised Learning
III. Reinforced learning
IV. Others…

Where is NN:

Almost all of them



What algorithm we have learned:

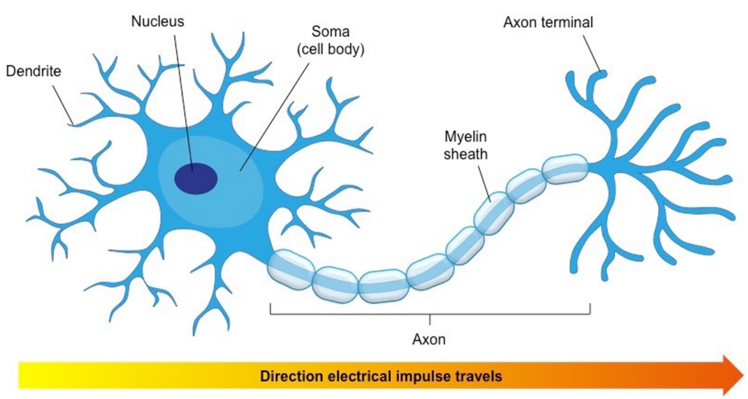


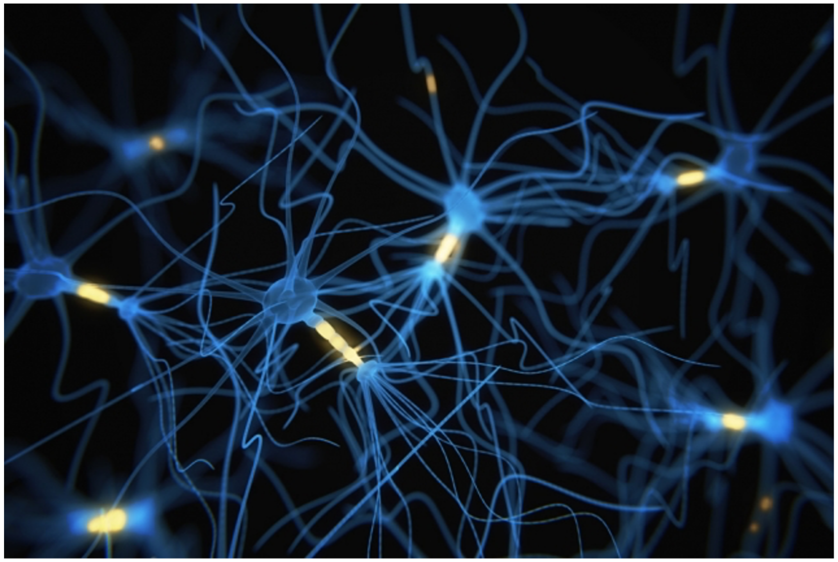
Neural Networks

What Neural Networks:

As the name suggest a network of neurons.

A Neural Network (artificial) is similar to neural networks in our brain. There are lots of connections between neuron, and they are constantly communicating. These connections are modeled by weights in an artificial neural network.



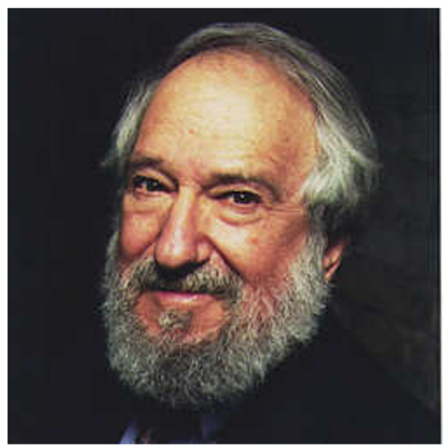


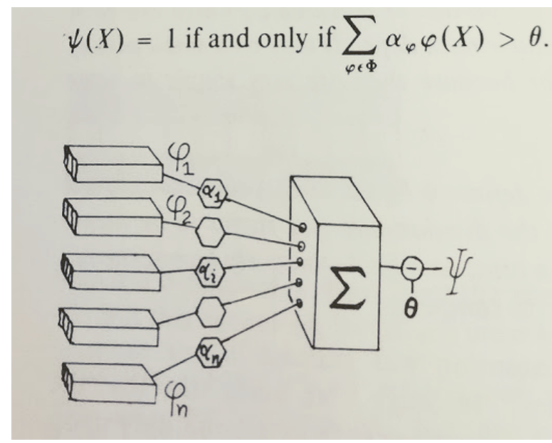
A bit of history and background:

The theory of Neural Network got its real attention after the famous publication of by Marvin Minsky and Seymour Papert (1969).

Yes it nothing new!

However the new computational power just let us appreciate its power very recently.

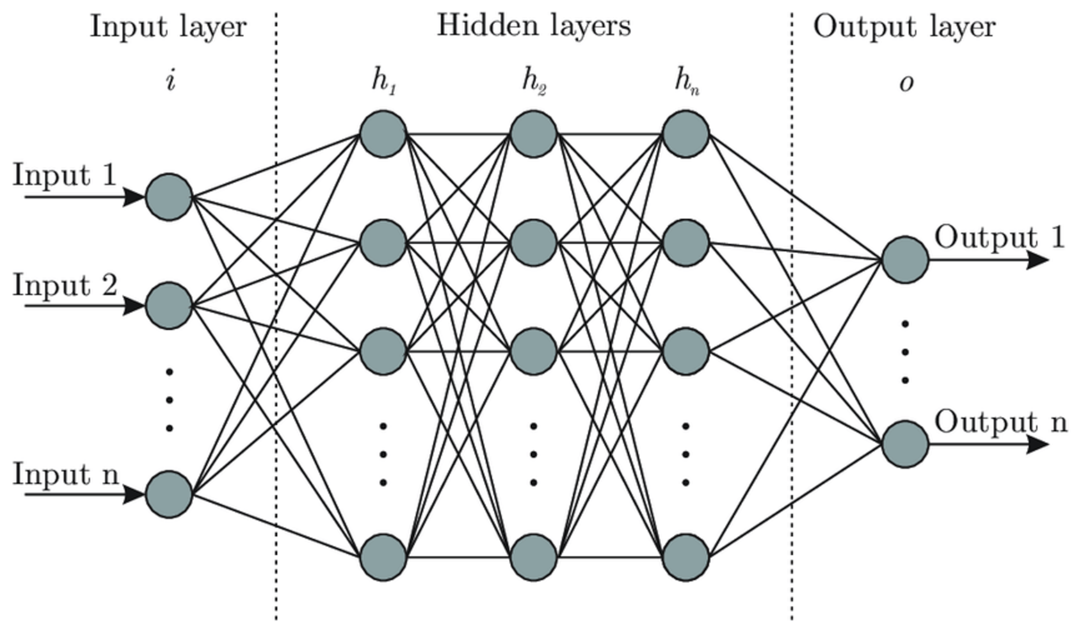




Artificial Neural Networks:

A fully connected layer is shown here, it consist of:

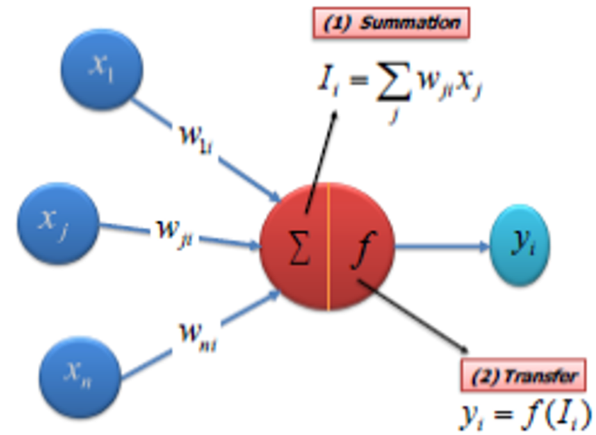
- Input layer
- Hidden layers
- Output layer

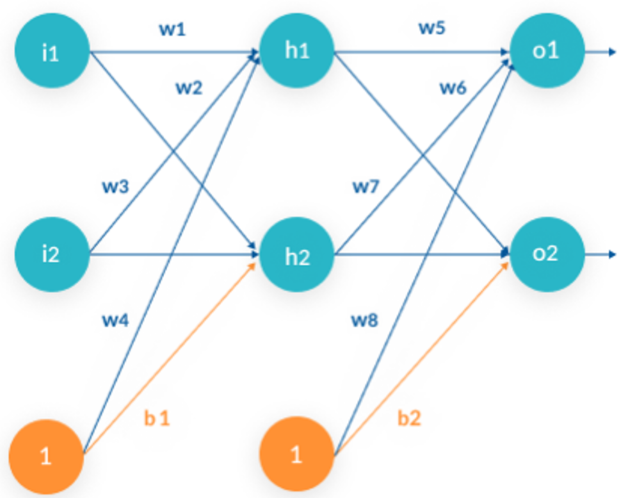


Artificial Neural Networks:

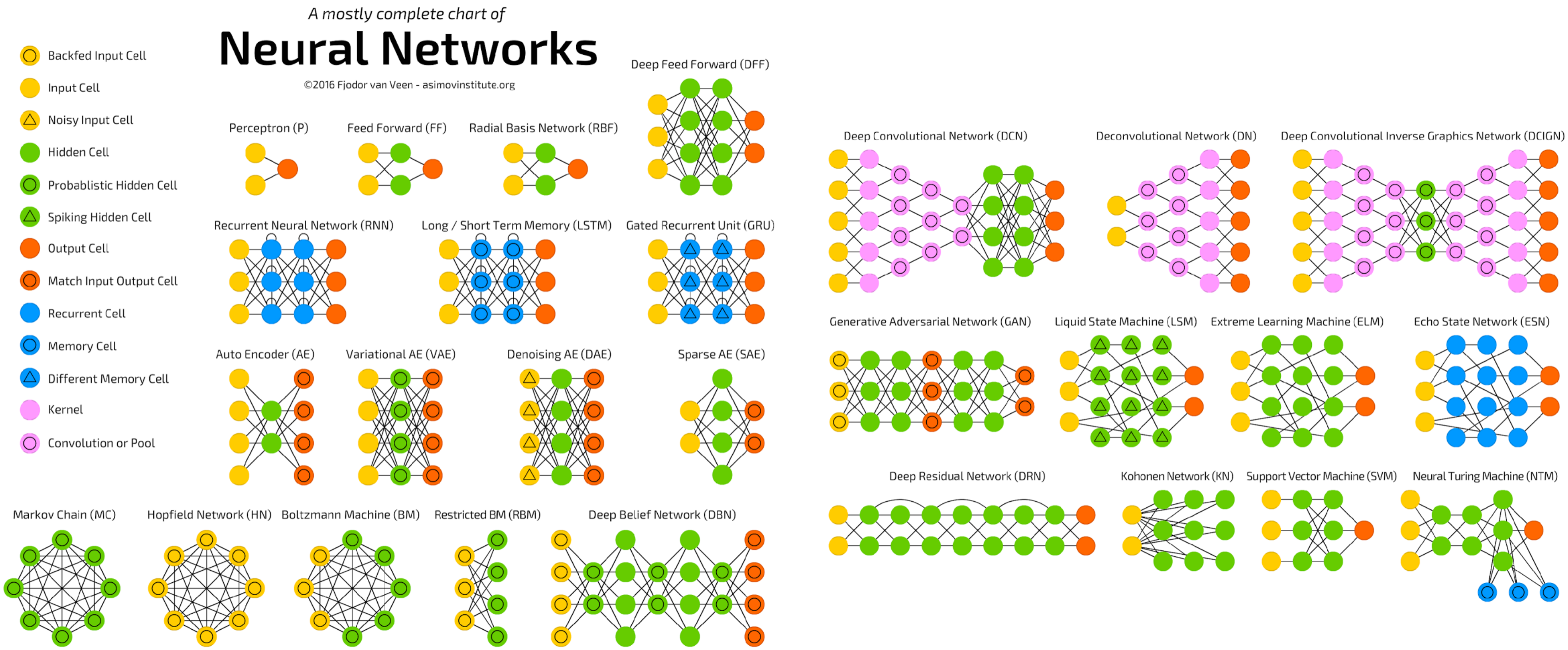
What else is there?

- Node (neuron values)
- Weights and Biases
- Activation functions





Different structures of Neural Networks:



Machine Learning revolutionized industries (or will) :

Here are some of ML applications:

I. Sale and retail, marketing, business development

II. Chatbots, revolutionized online services

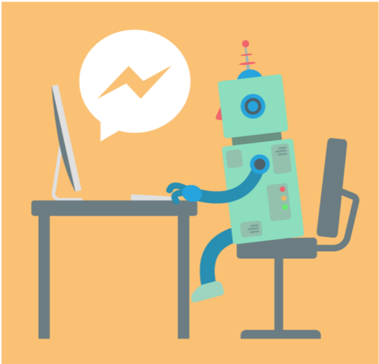
III. Manufacturing, by automation, replacing robots

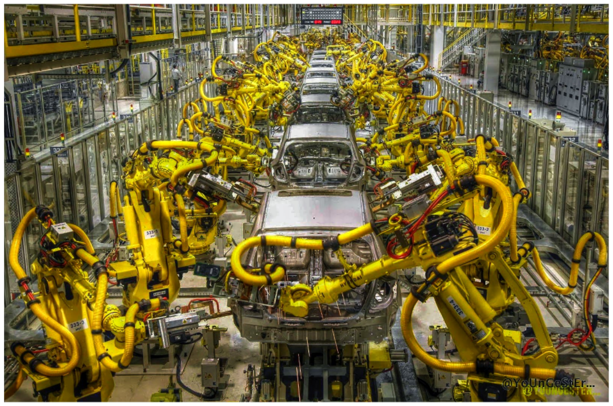
IV. Healthcare, X-rays and image processing, diagnose diseases (earlier than we do now),

V. Educations

VI. Real state and construction industry, (smart homes, automate construction)

VII. …





Applications of Neural Networks:

Look at some cool applications:

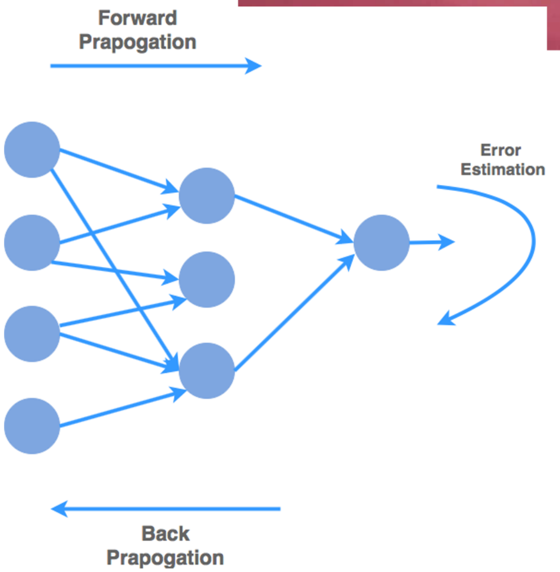
https://www.youtube.com/watch?v=kopoLzvh5jY

https://www.youtube.com/watch?v=LY7x2Ihqjmc
https://www.youtube.com/watch?v=vUgUeFu2Dcw

Intuition behind Neural Network: How it works

Remember the tennis player?

- Forward Propagation (feedforward)
- Backward Propagation



Intuition behind Neural Network: How it works

Forward Propagation:

- It is the algorithm that generate the output based on the inputs and other parameters.
- In this process the flow is only in one direction (forward), it is used to generate output, therefore it can not have backward flow (or otherwise it will create a loop)
- It is used in training process, and
- When the model is trained and weights are determined it will be also used to predict results.
- An it has 2 main step:
  - Pre-activation: It is a weighted sum of inputs
  - Activation: The Pre-activation is passed to a function

Intuition behind Neural Network: How it works

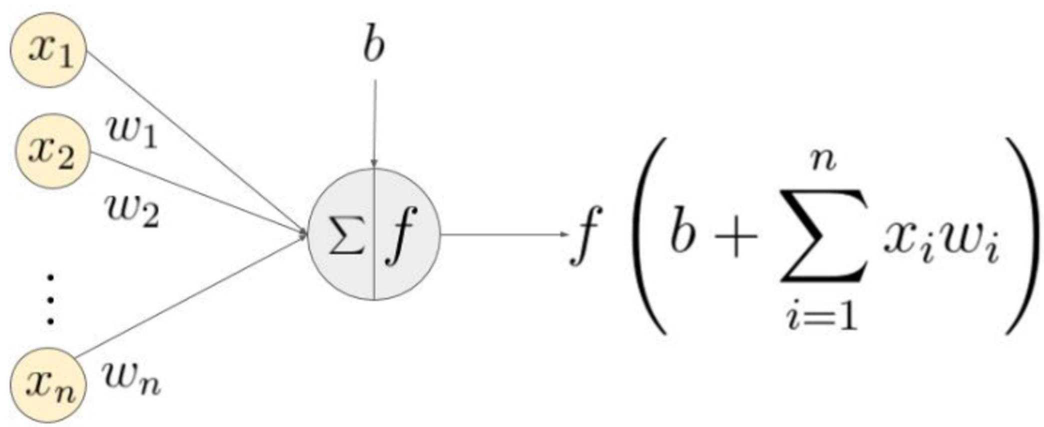
Forward Propagation:

Pre-activation: It is a weighted sum of inputs

$$h_1 = w_1 * x_1 + w_2 * x_2 + \cdots + w_n * x_n + b$$

Activation: The Pre-activation is passed to a function

$$z = f(h_1)$$



Intuition behind Neural Network: How it works

Forward Propagation simple example:

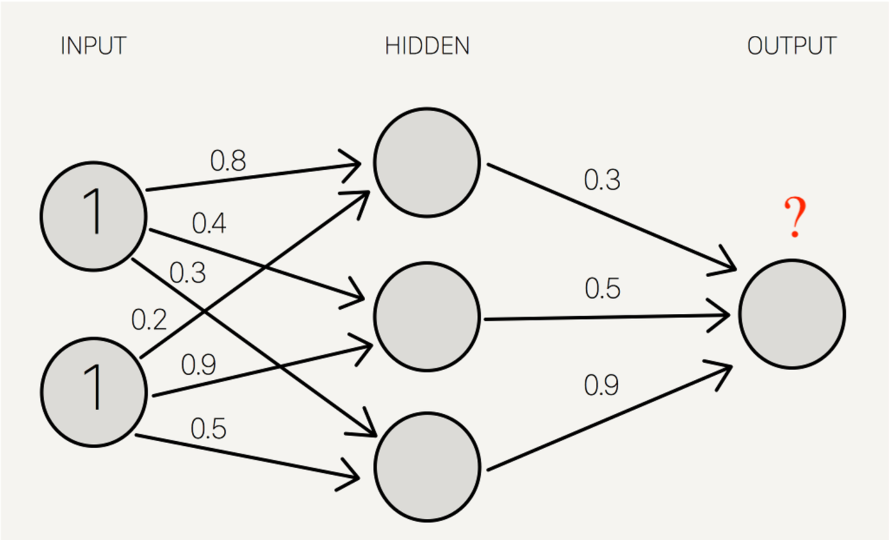
Use a bias of 0.5 for all nodes of Hidden layer and a bias of 0.8 for all bias in the output layer.

Use Tangent function as the activation function in the hidden layer

Use a pass through (linear) activation in the last layer, as follow

$$F(x) = x$$

(its like you basically don't apply a function).



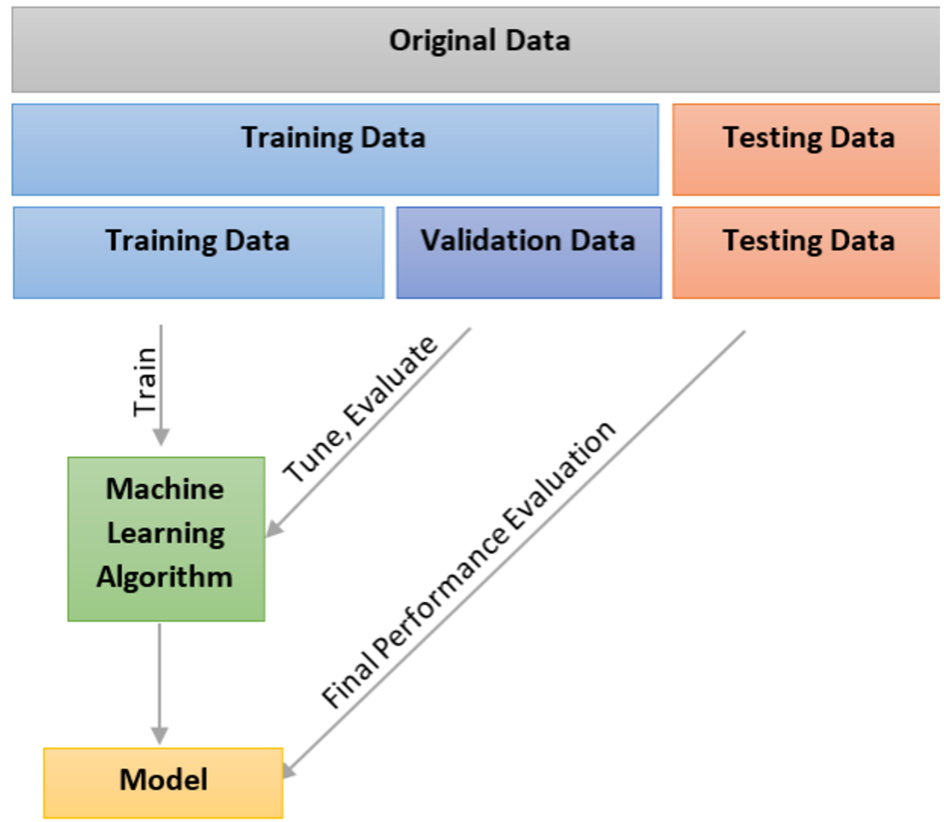
Python:

Install Keras TensorFlow

Other good library is Pytorch

Training vs Test sets:

1) Split dataset into Train and test sets
2) Train model on the training set
3) Tune model parameters on Validation set
4) Perform final evaluation on test set



Python:

For almost every model we do, we follow these: